

Future Outlook of Dumpsite Remediation Industry in India

- Presentation
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Solid Waste Rules 2025- Dumpsite Management

Solid Waste Rules 2025 – MSW Landfill Management

National Statistics on Dumpsite Remediation

Source: SBM
DASHBOARD

India | Status of Dumpsites Profile



No. of Dumpsites:2425 *

* Primarily dumpsites having legacy waste >= 1000 Tonne

Legacy Waste Qty.	Remediated Waste Qty	Waste to be Remediated
2,357	1,167 (49%)	1,191 (51%)
Area	Area Reclaimed	Area to be Reclaimed
14,881.28	6,239.06 (42%)	8,642 (58%)

* Waste Qty in Lakh Tonne

* Area in Acre

Remediation Status

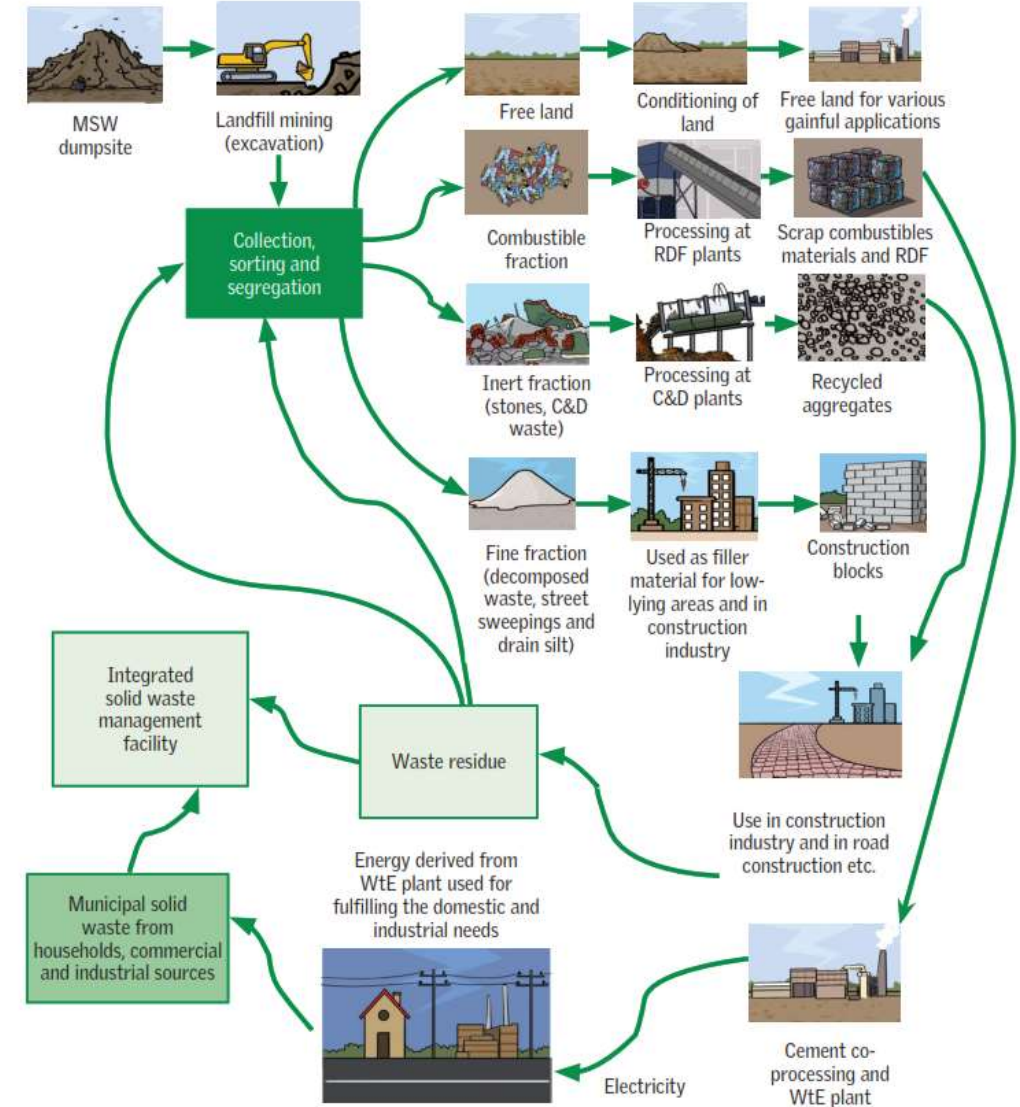
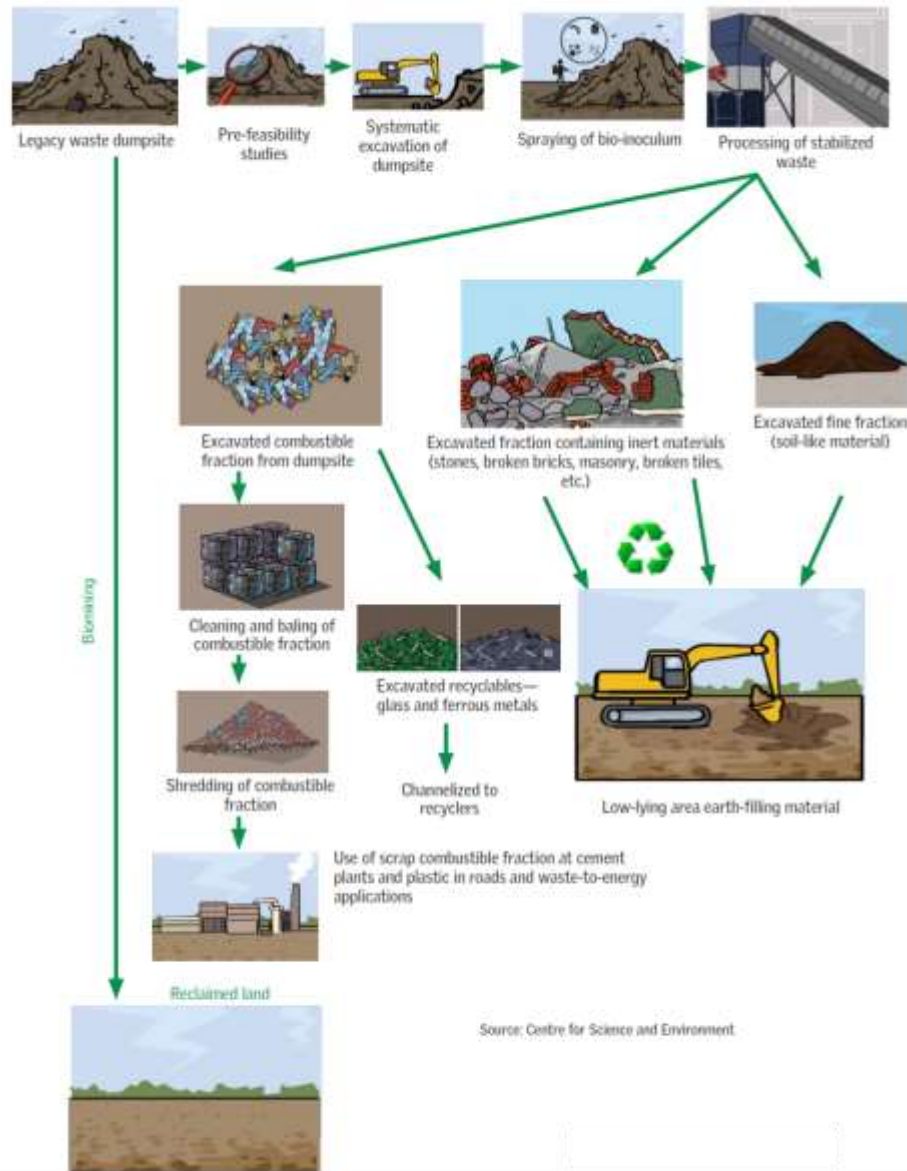
Remediation Completed (697)
No of Dumpsite

Legacy Waste Qty.	Remediated Waste Qty	Waste to be Remediated
684	684 (100%)	0 (0%)
Area	Area Reclaimed	Area to be Reclaimed
3,973.28	3,973.28 (100%)	0 (0%)

Remediation Approved & On-going (1182)
No of Dumpsite

Balance (546)
No of Dumpsite

Dumpsite Remediation and Circular Economy



Circular Economy Impact from Dumpsite Remediation in India

Direct Jobs Creation :

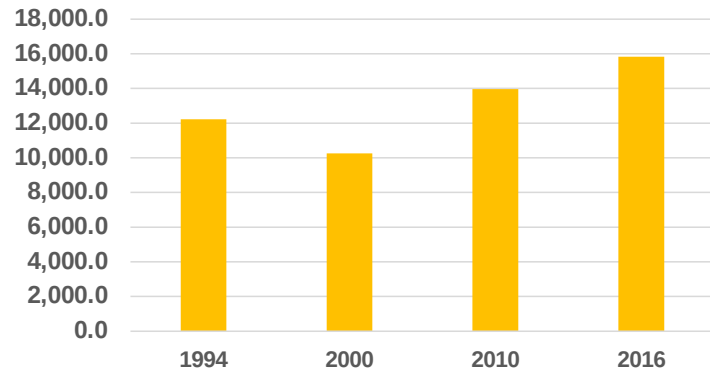
- As per estimates, dumpsites hold **2357 lac** tons of legacy waste.
- A **100 TPD** dumpsite remediation project has the potential to generate **66** man-days of job. Based upon this value, a total of **1.76 lac** human days of direct employment have been generated in the country from **1167** lac tons of legacy waste remediation.

Indirect Jobs Creation

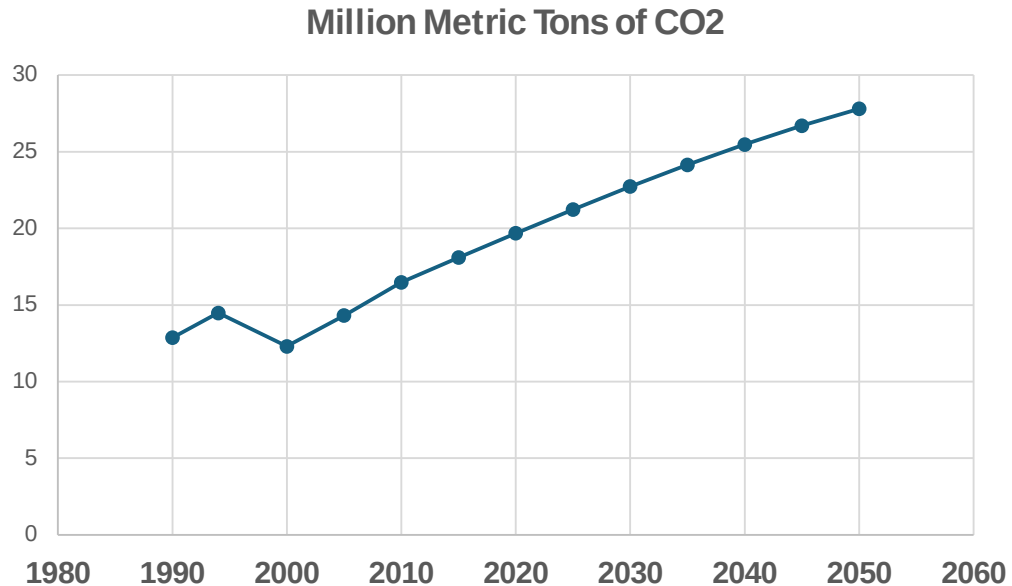
- As per estimates, dumpsites hold 2357 lac tons of legacy waste. Based on excavated statistics, the typical legacy waste has the following composition :
 - = **70% of Inerts (Soil); 15% of dry waste with 8% Plastics and other fractions**
- Taking the above values, the quantity of inert soil recovery potential is – **$2357 \times 70\% = 1649$ lacs tons**
- Taking the above values, the quantity of plastics recovery potential is – **188.56 lac tons**
- Additionally, the recovery potential of the other dry waste fractions is **188.56 lac tons**.
- The total inerts recovered from the dumpsite have the potential to construct 610 km of low-end roads. This may result in the generation of 39 lac human days of job @6400 human days per km of rural road construction.
- For every ton of plastic waste recycled, 10 jobs are created*. Taking this value, **9.42 lac** human days of employment have been made in the country.

Climate Benefits of Dumpsite Remediation

Methane Emissions from Solid Waste Disposal on Land



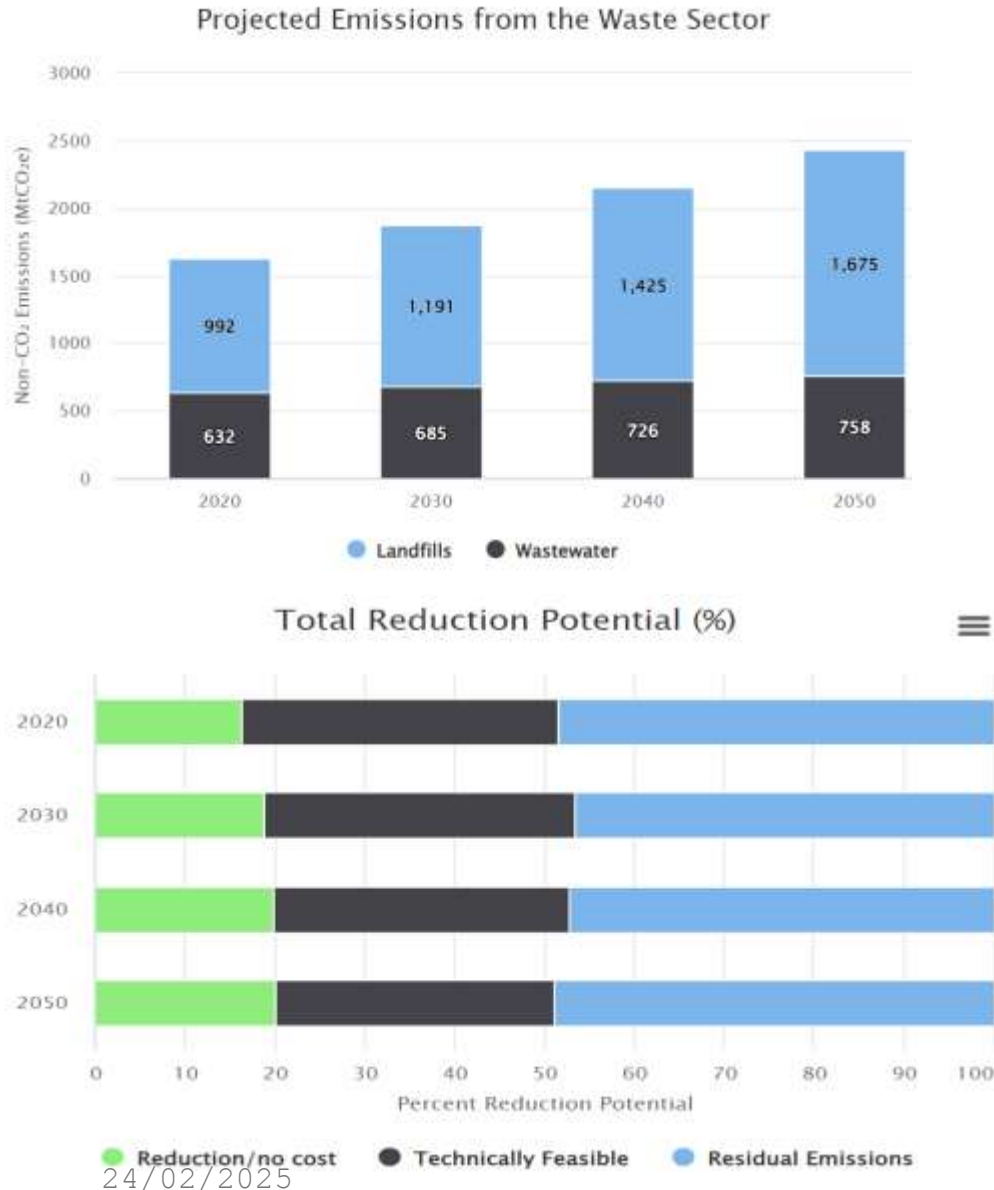
MSW Sector CO2 Emissions Reported by India



Projected Future MSW Sector CO2 Emissions (Global Methane Hub)

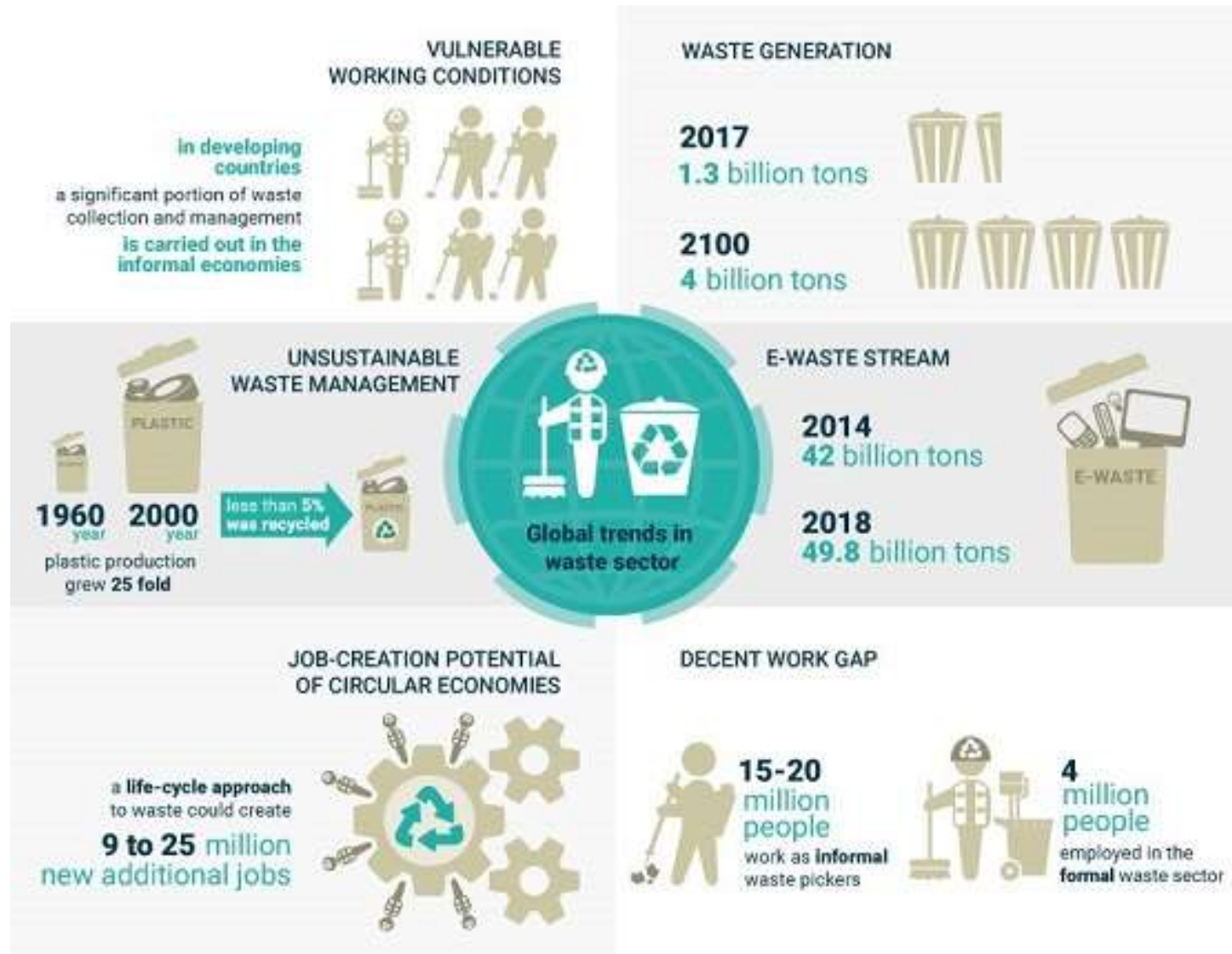
- The top left graph starkly illustrates the urgent need for action, showing that methane emissions from the solid waste sector are projected to continue their alarming increase during the forecast period from 1990 to 2050.
- The lower left graph illustrates a significant 22% change in emissions from 1994 to 2016, underscoring the urgent need for effective waste management strategies.
- Under the dumpsite remediation, 1176 lac tons of waste has been remediated.
- An estimated 2.43 million tons of methane emissions have been avoided by undertaking nationwide dumpsite remediation, assuming that each ton of solid waste disposed of in India could emit approximately 0.021 metric tons (21 kg CH₄/Ton) .
- waste generated in the country is associated with a 15% contribution of PM2.5 generation. The dumpsite remediation program has reduced it significantly.
- Land reclamation from dumpsite remediation has led to land parcel availability, which is the most considerable challenge for cities in expanding urban waste management systems.
- In 2023, Zigma, the largest biomining company in India, made a significant contribution to the environment by sending 5.7 lac tons of RDF, helping the cement industry reduce its CO2 emissions by up to 2415 tons per annum to date.

Future Global Business Outlook for dumpsite remediation



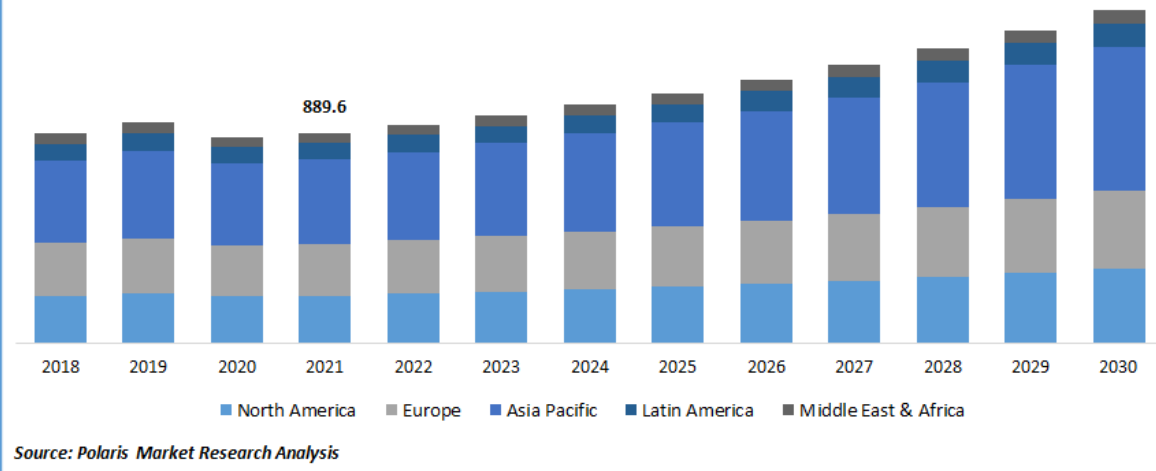
- The global landfill mining market has been experiencing steady growth. In 2023, the market was valued at approximately USD 386.4 million and is projected to reach USD 526.2 million by 2032, growing at a compound annual growth rate (CAGR) of 3.6% from 2024 to 2032.
- Between 2015 and 2030,, global waste-sector emissions are projected to increase 23% under a BAU scenario, reaching 1,905 MtCO₂e in 2030.
- Increases in population and per capita waste generation drive global waste emissions upward, but historical implementation of waste-related regulations and gas recovery and use has tempered this increase.
- Global waste-sector CH₄ and N₂O historical and projected emissions and the mitigation potential are estimated from the landfills and dumpsites
- Mitigation potential from the waste sector is estimated to be approximately 887 MtCO₂e in 2030. This mitigation potential is 53% of landfill; 47% of waste-sector emissions; and 23% of total global non-CO₂ mitigation potential in that year.

MAPPING GREEN DECENT JOBS IN THE GLOBAL WASTE SECTOR



Global Factors Driving Waste Management Jobs

Waste Management Market Size, By Region, 2018 - 2030
(USD Billion)

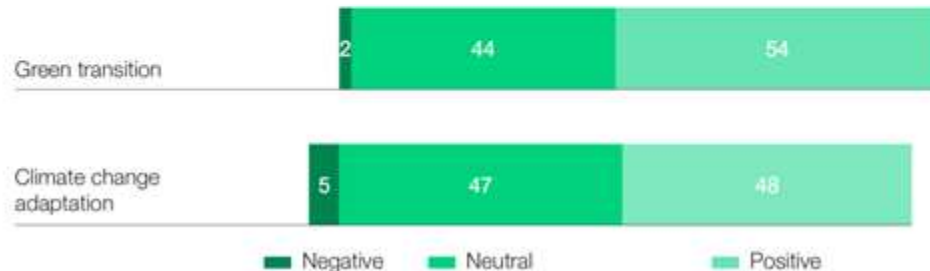


Future of Jobs

Green transition drives job growth



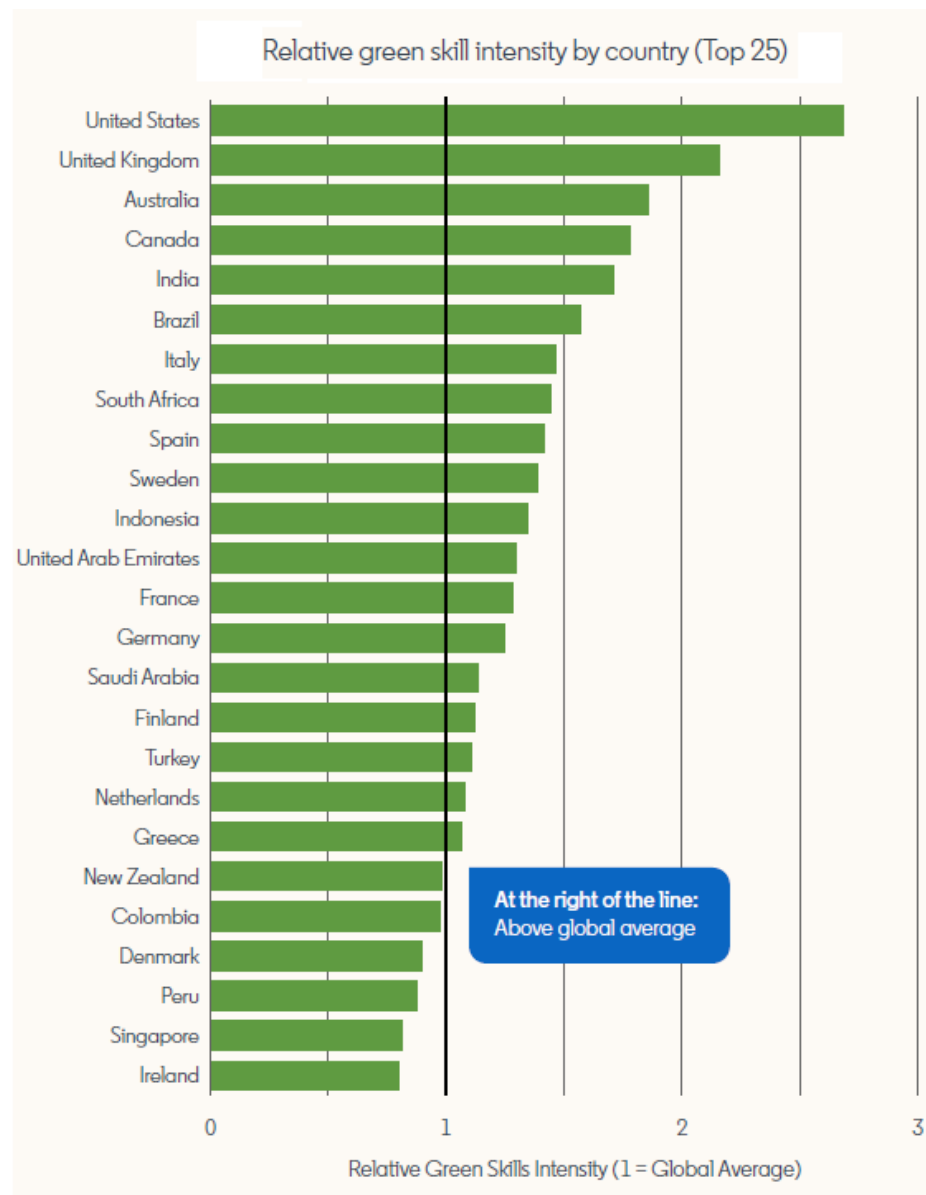
Expected impact of trends on jobs:



Number of Recycling Venture Capital Deals by Region, 2012 - 2021 YTD



Green Skills Intensity –India Position & Global Growth



Fastest-growing green skills (2016–2021)

Skill name	Green skill category	Skill growth
Sustainable fashion	Pollution Prevention	90.6%
Environmental services	Ecosystem Management	82.5%
Oil spill response	Environmental Remediation	80.4%
Climate	Ecosystem Management	68.7%
Sustainable Growth	Environmental Auditing	67.2%
Surface water	Ecosystem Management	64.5%
Occupational Safety and Health Advisor (OSHA)	Environmental Policy	57.9%
Sustainable business strategies	Pollution Prevention	56.6%
Solar systems	Renewable Energy Generation	55.5%
Sustainable landscapes	Ecosystem Management	52.9%

SWM RULES 2025 – DUMPSITE MANAGEMENT

- The local body must map and assess dumpsites and garbage vulnerable points for solid waste accumulation by March and October 2026, with online information.
- Legacy waste dumpsites must be bio-mined and bioremediation, with progress updates on a centralized online portal and local body website by the 30th of the first month of the next quarter.
- The local body requires project executing agencies to disclose proposed destinations for biomined solid waste, including fine soil, RDF, and C&D waste, and to test these for environmental safety, energy recovery, and processing standards.
- Local bodies must ensure environmentally sound biomining or bioremediation of legacy waste through a third party authorized by SPCB and file quarterly returns by the 30th of the following month.
- The State Pollution Control Board must conduct an annual audit on bio-mining and bio-remediation progress in dumpsites, with the report uploaded by December 31st of the following year.
- The concerned State Government or landowning agency may consider using the vacant land of a dumpsite after clearing waste/ legacy waste for the establishment of solid waste management infrastructure, subject to applicable regulations.

SWM RULES 2025 – MSW LANDFILL MANAGEMENT

- The sanitary/operational landfill is exclusively for non-recyclable and non-energy-recoverable dry waste and inerts, and wet or construction/demolition waste is not disposed of.
- The department must determine and notify the weight-based landfill user fee for urban/rural bodies, or third parties based on the quantity of unsegregated, recyclable, energy-recoverable, unprocessed, or construction waste.
- The operator charges landfill fees for unsegregated, mixed, recyclable, energy recoverable, unprocessed wet, construction, and demolition waste deposited by the local body or authorised third parties.
- The sanitary/operational landfill charges a higher user fee for unsegregated, recyclable, energy-recoverable, unprocessed biodegradable, construction, and demolition waste, which is deposited in a separate account by the local body for waste processing and waste management infrastructure development.
- Agencies operating sanitary/operational landfills must charge fees for inerts and allowed wastes, following CPCB guidelines and SPCB recommendations. Fees are determined by the UDD/concerned State/UT Government department.
- The State Pollution Control Board must audit landfill operations for environmentally sound waste disposal, with audit reports due by December 31st of the following year.
- The sanitary/operational landfill operator must register, file annual returns, and comply with environmental guidelines set by the CPCB. They must also submit quarterly reports to the District Magistrate to ensure compliance.

Golden Rules for Post-Dumpsite Remediation

- Dumpsite remediation of sites located near riverbanks should not be undertaken. A capping and Closure approach should be adopted.
- Reclaimed land should be used to construct new landfills and processing plants for future requirements. Inert recovered from dumpsites should also be used for construction purposes, as it reduces new landfill construction costs by 25% - 30%.
- The company undertaking dumpsite remediation should explore business continuity by undertaking waste processing and disposal business, as compared to simple remediation and handover of land.
- Reclaimed land should be mandatorily used for meeting future SWM expansion for the city.